Venkata Mangalampeta

510-972-4348 vmangalampeta@ucsd.edu linkedin.com/in/venkata-mangalampeta

Education

University of California – San Diego **Graduation: Jun 2027**
B.S. in Data Science *La Jolla, CA*

Relevant Coursework: Data Structures & Algorithms, Operating Systems & Computer Systems, Machine Learning, Probability & Statistics, Software Engineering

Experience

IBM **Oct 2026 – Dec 2026**
Data Engineer Intern *Baton Rouge, LA*

- Incoming Data Engineer Intern - Fall 2026

Interpret Lab **Jan 2025 – Sep 2025**
Software Engineering Intern *Santa Cruz, CA*

- Developed fault-tolerant Python data services in an Agile sprint workflow with schema validation and automated backfills, enabling safe re-runs and consistent versioned datasets for downstream consumers.
- Profiled and debugged pipeline runs using distributed traces, metrics, and system diagnostics; reduced runtime variance and improved production stability across repeated jobs.
- Integrated structured logging and automated regression checks into the continuous delivery pipeline, catching failures before releases and accelerating on-call triage.

Soshal **Mar 2024 – Nov 2024**
Software Developer *Remote*

- Built and shipped REST API backend features for a production SaaS application serving 10,000+ users, integrating with a relational database and maintaining full CI/CD pipelines via Git for versioned releases.
- Improved reliability for high-traffic events (500+ attendees) by adding Redis caching, optimizing SQL queries, and hardening API behavior with timeouts, retries, and graceful error handling.
- Containerized services with Docker and deployed on AWS IaaS infrastructure, adding health checks, structured logs, and metrics to support production debugging and observability.

Tech4Good **Sep 2024 – Jan 2025**
Software & Machine Learning Intern *Santa Cruz, CA*

- Built Python preprocessing and indexing pipelines for large corpora powering a generative AI retrieval system; improved search quality by reducing noisy chunks through chunking, metadata tagging, and query cleanup.
- Applied prompt engineering and RAG retrieval tuning (embeddings, vector search, reranking) to cut incorrect answers over time; built eval harnesses with gold queries and regression suites to detect quality drift.
- Implemented structured logging with trace IDs and strict input validation to isolate parsing failures and edge cases across release cycles.

Projects

CiteCraft — AI Workflow Copilot | Python, FastAPI, PostgreSQL, Docker, AWS **Jun. 2025**

- Built an end-to-end generative AI copilot using RAG with citation-backed answers, hybrid retrieval, and schema-validated outputs; applied prompt engineering to improve grounding and response reliability.
- Designed a document ingestion pipeline (PDF parsing, chunking, embeddings, indexing) and evaluation scripts measuring Recall@K/MRR; added a feedback loop to iteratively expand the eval set and drive quality gains.

TransitETA: Real-Time ETA Forecasting | Python, FastAPI, SQL, Docker, AWS **Jan. 2025**

- Designed real-time ingestion from transit and weather feeds; generated cleaned features and automated daily refresh jobs to keep ETA inputs current for a production ML forecasting service.
- Built a FastAPI backend with Redis caching and optimized SQL queries; containerized with Docker and deployed on AWS with structured logging and health checks for production readiness.

Technical Skills

Languages: Python, JavaScript, SQL
Backend: FastAPI, REST API design, request lifecycle, validation, error handling, service integration
Data/DB: PostgreSQL, schema design, query optimization, indexes/joins, Redis
Cloud/DevOps: AWS (EC2/S3/IAM), Docker, Linux, Git/GitHub, Jenkins, CI/CD pipelines, pytest
Systems: Data structures/algorithms, OOP, Agile development, system debugging, data quality/validation
AI/ML: Generative AI, prompt engineering, RAG pipelines, embeddings/vector search, ML training/eval